

Discussion 4: The Lorentz Transformations – Solutions

Names:

Section:

In lecture this week, you saw how the Lorentz transformations (the relativistic versions of the Galilean transformations we studied in Week 1 and Week 3) can be derived from the two postulates of relativity, plus the assumption that coordinate transformations should be linear. That derivation was quick, but rather formal and mathematical. In this discussion, you will see how the same Lorentz transformations can also be derived by thinking carefully about the three fundamental effects of relativity that you studied in Week 2. You will also see how you can work backwards to recover the three fundamental effects from the Lorentz transformations. This discussion follows Morin Sec. 2.1 closely, so feel free to refer there for inspiration, but note that our notation here is slightly different so you can't copy the results directly.

Throughout this discussion, we will work in our usual setup where (unless otherwise specified) frame S is stationary, frame S' is moving in the $+x$ direction at speed v , and the two coordinate systems have synchronized their clocks to $t = 0$ at the origin so that we can drop the Δ 's.

Part 1: Deriving the Lorentz transformations

The most general linear transformation between (x, t) and (x', t') is

$$t = At' + Bx', \quad (1)$$

$$x = Ct' + Dx', \quad (2)$$

where A , B , C , and D are constants (i.e. independent of x, t, x', t') which can depend on v . We will use the three fundamental effects along with the definition of frame S' to solve for these four constants. The quickest way to the Lorentz transformations is to zero out each coordinate independently, and perform a suitable measurement combined with a suitable fundamental effect. This is what Morin Section 2.1 does, but it requires a considerable amount of frame-switching. Instead, we will perform a more physically transparent derivation, staying entirely in one frame, but which requires you to solve some equations for the coefficients.

Working entirely in frame S :

- 1.1. Use the fact that the S' clock is running slow compared to S to show that $A = \gamma$.

We can look at a fixed clock in S' at $x' = 0$ to find $t = At'$. Time runs slowly by a factor γ in S' , so $A = \gamma$.

- 1.2. Use the fact that lengths in S' as seen in S are contracted to show that $\frac{B}{A} = \frac{D-1/\gamma}{C}$. (Note that you will have to use both of the Lorentz transformation equations.)

A length in S' is a distance L , which we can imagine measuring at $t = 0$ in S . In other words, we compare the position of $x' = 0$ and $x' = L$ at $t = 0$. Since $x' = 0$ corresponds to $x = 0$ at

$t = 0$ (the origins of S and S' coincide), we only care about the value of x corresponding to $x' = L$, which by length contraction must be L/γ . We then have to solve

$$\begin{aligned} 0 &= At' + BL \\ \frac{L}{\gamma} &= Ct' + DL \end{aligned}$$

The first equation tells us $t' = -BL/A$, and plugging into the second yields

$$\frac{L}{\gamma} = -\frac{CBL}{A} + DL \implies \frac{B}{A} = \frac{D - 1/\gamma}{C}.$$

- 1.3. Use the fact that the origin of S' is moving to the right at speed v compared to the origin of S to show $C/A = v$.

The origin of S' is $x' = 0$, which must have position $x = vt$ in S . Setting $x' = 0$ and $x = vt$ in both Lorentz transformation equations gives $t = At'$ and $vt = Ct'$, and dividing gives $C/A = v$.

- 1.4. Use the rear-clock-ahead effect on the moving clocks in S' to show that $B/A = v/c^2$.

Consider the clocks in S' at $x' = -L$ and $x' = 0$, observed at $t = 0$ in S' . The $x' = 0$ clock must read $t' = 0$ since the origins are synchronized, and the $x' = -L$ clock is the rear clock, so it reads $t' = Lv/c^2$. Plugging $t' = Lv/c^2$, $x' = -L$, and $t = 0$ into the first Lorentz transformation equation gives $0 = ALv/c^2 - BL$, or $B/A = v/c^2$.

- 1.5. Solve for A , B , C , and D using the above relations.

Problem 1.1 gives $A = \gamma$, problem 1.4 gives $B = \gamma v/c^2$, and problem 1.3 gives $C = \gamma v$. Using these results in problem 2 gives $\frac{v}{c^2} = \frac{D - 1/\gamma}{\gamma v}$. Rearranging,

$$D = \frac{1}{\gamma} + \frac{\gamma v^2}{c^2} = \frac{c^2 + \gamma^2 v^2}{\gamma c^2}.$$

The numerator is $c^2 + \frac{v^2}{1-v^2/c^2} = \frac{c^4}{c^2-v^2} = c^2\gamma^2$, so $D = \frac{c^2\gamma^2}{\gamma c^2} = \gamma$.

We now have the correct form of the Lorentz transformation equations:

$$t = \gamma(t' + \frac{v}{c^2}x') \quad (3)$$

$$x = \gamma(vt' + x') \quad (4)$$

Make sure your values of A , B , C , and D agree with these before moving on!

Part 2: Lorentz transformations in matrix form

The fact that the Lorentz transformations are linear means that we can represent them with a matrix, just as we did for Galilean transformations last week.

- 2.1. Write the matrix equation transforming the spacetime vector (t', x') in S' to the corresponding vector (t, x) in S .

We can just strip off the coefficients in the Lorentz transformation equations and collect them in a matrix:
$$\begin{pmatrix} t \\ x \end{pmatrix} = \begin{pmatrix} \gamma & \frac{\gamma v}{c^2} \\ \gamma v & \gamma \end{pmatrix} \begin{pmatrix} t' \\ x' \end{pmatrix}$$

- 2.2. We can make the Lorentz transformations a lot more symmetrical by writing all quantities in units of the speed of light. Let's define $\beta = v/c$ (note $|\beta| < 1$ because $|v| < c$), and write ct for the time coordinate (in other words, if t is 1 second, then ct is 1 light-second, which has units of distance). Show that the Lorentz transformations can be written in matrix form as

$$\begin{pmatrix} ct \\ x \end{pmatrix} = \begin{pmatrix} \gamma & \gamma\beta \\ \gamma\beta & \gamma \end{pmatrix} \begin{pmatrix} ct' \\ x' \end{pmatrix}. \quad (5)$$

Changing variables from t and t' to ct and ct' in the Lorentz transformation equations means multiplying the t equation by c and factoring out a c from the x equation:

$$\begin{aligned} ct &= \gamma(ct') + \gamma\frac{v}{c}x' \\ x &= \gamma\frac{v}{c}(ct') + \gamma x' \end{aligned}$$

Replacing v/c with β and putting the coefficients into a matrix gives the desired form.

- 2.3. Solve the linear equations for (ct', x') in terms of (ct, x) to find the inverse Lorentz transformation, in other words the matrix which transforms S to S' . Once you have it, explain how you could have written down this answer immediately without doing any algebra.

Multiplying the x equation by β and subtracting kills the x' terms and gives $ct - \beta x = (\gamma - \gamma\beta^2)ct'$. We can simplify the right-hand side further with $1 - \beta^2 = 1 - v^2/c^2 = 1/\gamma^2$. So multiplying both sides by γ gives $ct' = \gamma(ct - \beta x)$. Similarly, multiplying the ct equation by β and subtracting kills the ct' terms and gives $\beta ct - x = (\gamma\beta^2 - \gamma)x' = \gamma(\beta^2 - 1)x' = -x'/\gamma$. Multiplying both sides by $-\gamma$ gives $x' = \gamma x - \gamma\beta ct$. Thus, the inverse Lorentz transformation matrix is
$$\begin{pmatrix} \gamma & -\gamma\beta \\ -\gamma\beta & \gamma \end{pmatrix}$$
. This is simply the original Lorentz transformation matrix with the sign of the velocity flipped, $v \rightarrow -v$, which preserves γ but takes $\beta \rightarrow -\beta$. We could have written this down immediately by looking at S from the perspective of S' , which moves with relative velocity $-v$.

- 2.4. We've been neglecting the roles of the coordinates y and z in the transverse directions, because they don't change when the relative motion between S and S' happens only in the x direction: $y' = y$ and $z' = z$. Restore these coordinates to find the 4×4 matrix which transforms (ct', x', y', z') to (ct, x, y, z) .

The third row of the matrix is the equation for y , and the fourth row is the equation for z . Since $y = y'$, the entries of the third row must be the vector $(0, 0, 1, 0)$. Likewise, the entries of the fourth row are the vector $(0, 0, 0, 1)$. Finally, the entries in the upper-right 2×2 corner of the matrix would be the coefficients of y' and z' in the equations for ct and x . But there

are no such terms, so these entries are just zero. We thus have

$$\Lambda_x(v) = \begin{pmatrix} \gamma & \gamma\beta & 0 & 0 \\ \gamma\beta & \gamma & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix}$$

- 2.5. Without doing any additional algebra, find the matrices which transform (ct', x', y', z') to (ct, x, y, z) in the cases where S' is moving in the positive y -direction or the positive z -direction instead of the positive x -direction.

We just need to replace the entries that involve the x and x' coordinates with either y, y' or z, z' . This is accomplished by rearranging the 1's on the diagonal and putting the γ in the third or fourth entry on the diagonal, and moving the $\gamma\beta$ terms to the appropriate row and column. This gives

$$\Lambda_y(v) = \begin{pmatrix} \gamma & 0 & \gamma\beta & 0 \\ 0 & 1 & 0 & 0 \\ \gamma\beta & 0 & \gamma & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix}, \quad \Lambda_z(v) = \begin{pmatrix} \gamma & 0 & 0 & \gamma\beta \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ \gamma\beta & 0 & 0 & \gamma \end{pmatrix}$$

You can check that these are correct by performing the matrix multiplication by (ct', x', y', z') . For example, multiplying by Λ_y gives the equations

$$\begin{aligned} t &= \gamma ct' + \gamma\beta y' \\ x &= x' \\ y &= \gamma\beta ct' + \gamma y' \\ z &= z' \end{aligned}$$

We will close this part by introducing some notation which you will use for the remainder of the course. The object (ct, x, y, z) is called a *spacetime 4-vector* (because it combines space and time, and has 4 components). It is often given the name x^μ , where the superscript is a Greek letter, and the indices start from zero: $x^0 = ct, x^1 = x, x^2 = y, x^3 = z$. The Lorentz transformation matrix is traditionally called Λ , and in index notation (using the Einstein summation convention) the Lorentz transformations are written as

$$x^\mu = \Lambda^\mu_\nu x'^\nu. \tag{6}$$

Lorentz transformations are often referred to as “boosts,” and γ is also called the “boost factor.”

Part 3: Recovering the fundamental effects

The Lorentz transformations encapsulate the *entire* content of the theory of special relativity. This means that all of the relativistic phenomena we have studied thus far, as well as more that we will see later this semester, can be derived directly from the Lorentz transformations without having to

resort to light clocks, trains, and so on. On the one hand, this reduces the quantity of formulae you have to remember! On the other hand, having a physical picture in mind is *very* helpful for keeping the factors of γ straight, so you are encouraged to use both techniques as you solve problems in this course.

- 3.1. Explain how to derive the rear-clock-ahead effect from the first Lorentz transformation equation, $t = \gamma(t' + \frac{v}{c^2}x')$.

As in Part 1, problem 1.4, we measure the clocks in S' at $t = 0$. Rearranging gives $t' = -\frac{v}{c^2}x'$. In other words, as we sit in S and scan down the negative x' -axis of S' , each clock we see is ahead of the one at the origin ($x' = 0$) by an amount $x'v/c^2$.

- 3.2. Set $x' = 0$ in the first Lorentz transformation equation to recover the time dilation effect, $t = \gamma t'$. Note that you can also set $x = 0$ in the inverse Lorentz transformations to find $t' = \gamma t$; S sees the clocks in S' run slow, and vice versa.

By setting $x' = 0$, we are measuring time elapsed on the clock at the origin of S' . The first Lorentz transformation equation gives $t = \gamma t'$, so time elapsed in S is longer than proper time in S' by γ .

- 3.3. Set $t' = 0$ in the second Lorentz transformation equation, $x = \gamma(vt' + x')$, to recover the length contraction effect. In this context, which frame is doing the measuring, and which (x or x') is the proper length? As in 3.2, you can use the inverse transformations to perform the measurement in the other frame.

By setting $t' = 0$, we are measuring simultaneous events in S' (not S !), and so x' is the length as measured in S' . The equation gives $x = \gamma x'$, or $x' = x/\gamma$, so an object with proper length x in S has contracted length x/γ in S' .

- 3.4. Now set $x' = 0$ in both Lorentz transformation equations to obtain $t = \gamma t'$ and $x = \gamma vt'$. Interpret this result.

These equations just tell us how the origin of S' moves in S . We already know about $t = \gamma t'$ from 3.2, so we plug this in to find $x = vt$: the origin of S' moves with constant velocity v in S .

- 3.5. Finally, revisit Part 2 of Discussion 1 using Lorentz transformations. Specifically, use the Lorentz transformation equations to derive the expressions for t_F and t_B for the times as measured in S that light rays emitted from the middle of a train in S' hits the front and back of the train. You will actually need to use the Lorentz transformations twice: once to relate the length of the train L_S as measured in S to its proper length in S' (this is just question 3.3 above), and again to relate the time of arrival of the light in the two frames. Note though that no extraneous information is required: you don't have to think about gap-closing times or anything like that, it's all contained in the Lorentz transformations as long as you're careful!

Let the proper length of the train in S' be L' . From 3.3 (but applied to the inverse transformations), we have $L = L'/\gamma$. Define the middle of the train to be the origin of S' , which coincides with the origin of S at $t = t' = 0$, when the light is emitted. In S' , the train clocks read the same time when the light hits, $t' = \frac{L'}{2c}$, and these clocks are located at $x'_B = -L'/2$

and $x'_F = L'/2$. Now, we can apply the first Lorentz transformation equation directly, to find the times t_B and t_F that the clocks in S read:

$$t_B = \gamma t' + \gamma \frac{v}{c^2} \left(-\frac{L'}{2} \right) = \gamma \left(\frac{L'}{2c} - \frac{vL'}{2c^2} \right) = \frac{\gamma L'}{2c} \left(1 - \frac{v}{c} \right) = \frac{\gamma^2 L}{2c} \left(\frac{c-v}{c} \right) = \frac{L}{2(c+v)},$$
$$t_F = \gamma t' + \gamma \frac{v}{c^2} \left(\frac{L'}{2} \right) = \gamma \left(\frac{L'}{2c} + \frac{vL'}{2c^2} \right) = \frac{\gamma L'}{2c} \left(1 + \frac{v}{c} \right) = \frac{\gamma^2 L}{2c} \left(\frac{c+v}{c} \right) = \frac{L}{2(c-v)}.$$

Note that the beauty of the Lorentz transformations is that we never had to find the locations x_F and x_B of the front and back of the train in S in order to solve the problem!